

Project 2 — One-Time Pad: Bob's Decryption

This notebook implements Bob's portion of the one-time-pad exercise.

Bob:

1. Reads the ciphertext from a hexadecimal text file.
2. Reads the key from a hexadecimal text file.
3. Converts both values to bytes.
4. XORs the ciphertext with the key.
5. Decodes and displays the recovered plaintext.

```
In [4]: from pathlib import Path
```

Functions: Get ciphertext & key

The following two code blocks are the XOR function `xor_bytes()` which contains the exception handler which evaluates if the ciphertext and key are non-matching in length, and the `read_hex_file()` to read in the hexadecimal content and return the bytecode.

```
In [5]: def xor_bytes(data: bytes, key: bytes) -> bytes:
        if len(data) != len(key):
            raise ValueError("Ciphertext and key must have the same length.")

        return bytes(d ^ k for d, k in zip(data, key))
```

```
In [6]: def read_hex_file(filename: str) -> bytes:
        hex_data = Path(filename).read_text(
            encoding="ASCII"
        ).strip()

        return bytes.fromhex(hex_data)
```

Bob's Decrypt & Read

This section will read in the ciphertext via and the key from hexadecimal encoding and XOR the stream to decrypt and display Alice's message. `read_hex_file` and decodes the XOR'ed hexadecimal to ASCII; it will then output the following proofs:

- ciphertext in hexadecimal
- key in hexadecimal
- Alice's plaintext

```
In [7]: ciphertext = read_hex_file("ciphertext.txt")
key = read_hex_file("key.txt")

plaintext_bytes = xor_bytes(ciphertext, key)
plaintext = plaintext_bytes.decode("ascii")

print("Ciphertext hex:", ciphertext.hex())
print("Key hex:      ", key.hex())
print("Plaintext:    ", plaintext)
```

```
Ciphertext hex: 488e90bfc9f39a9b86766d85c296ce361502fa959d06c07b05b5
Key hex:      05f7bc9fa69bbaf6ff561aeda3e2ee57356493fbf826a41a7c9b
Plaintext:    My, oh my what a fine day.
```

```
In [ ]:
```